Graphs of Sine and Cosine Functions

Section 1: Amplitude and Vertical Shift

1 Find the amplitude and midline equation of $y = -4\cos(x) + 3$.

a. Amplitude $= 4$, Midline: $y = 3$

b. Amplitude $= -4$, Midline: $y = 3$

c. Amplitude $= 4$, Midline: $y = -3$

d. Amplitude $= 8$, Midline: $y = 3$

2 What are the maximum and minimum values of $f(x) = 5\sin(x) - 2$?

a. Maximum $= 3$, Minimum $= -7$

b. Maximum $= 5$, Minimum $= -5$

c. Maximum $= 7$, Minimum $= -3$

d. Maximum $= 3$, Minimum $= -3$

3 A wave oscillates between a maximum of 12 and a minimum of 4. Find its amplitude.

a. 4

b. 8

c. 6

d. 16

Section 2: Period and Frequency

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4 Find the period of the function $y = 3\sin(4x)$.

a. $\frac{\pi}{2}$

b. π

c. 2π

d. $\frac{\pi}{4}$

5 Find the frequency (cycles per unit of x) of $y = \cos\left(\frac{\pi}{3}x\right)$.

a. 1/6

b. 6

c. $\pi/3$

d. 3

6 Write a sine function with amplitude 3 and period 4π.

a. $y = 3\sin\left(\frac{1}{2}x\right)$

b. $y = 3\sin(4x)$

c. $y = 3\sin(2x)$

d. $y = 4\sin\left(\frac{1}{3}x\right)$

Section 3: Phase Shift (Horizontal Shift)

7 Find the phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.

a. $\frac{\pi}{6}$ to the right

b. $\frac{\pi}{2}$ to the right

c. $\frac{\pi}{6}$ to the left

d. $\frac{3\pi}{2}$ to the right

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8 At what value of x does the first cycle start for $y = \cos(2x + \pi)$?

a. $-\frac{\pi}{2}$

b. π

c. $-\pi$

d. $\frac{\pi}{2}$

9 Which transformation shifts the graph of $y = \cos x$ to become identical to $y = \sin x$?

a. Shift right by $\frac{\pi}{2}$

b. Shift left by $\frac{\pi}{2}$

c. Reflect across the x-axis

d. Shift up by 1

Section 4: Modeling Periodic Phenomena

10 The temperature in a greenhouse follows $T(t) = 15\sin\left(\frac{\pi}{12}(t - 6)\right) + 65$, where t is hours after midnight. What is the maximum temperature?

a. 80°F

b. 65°F

c. 75°F

d. 95°F

11 In the greenhouse model above, at what time of day (t) does the maximum temperature occur?

a. $t = 12$ (12:00 PM / Noon)

b. $t = 6$ (6:00 AM)

c. $t = 18$ (6:00 PM)

d. $t = 24$ (Midnight)

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12 A Ferris wheel of radius 30 ft has its lowest point 5 ft off the ground and completes one rotation every 60 seconds. What is the midline height?

a. 35 ft

b. 30 ft

c. 65 ft

d. 40 ft

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